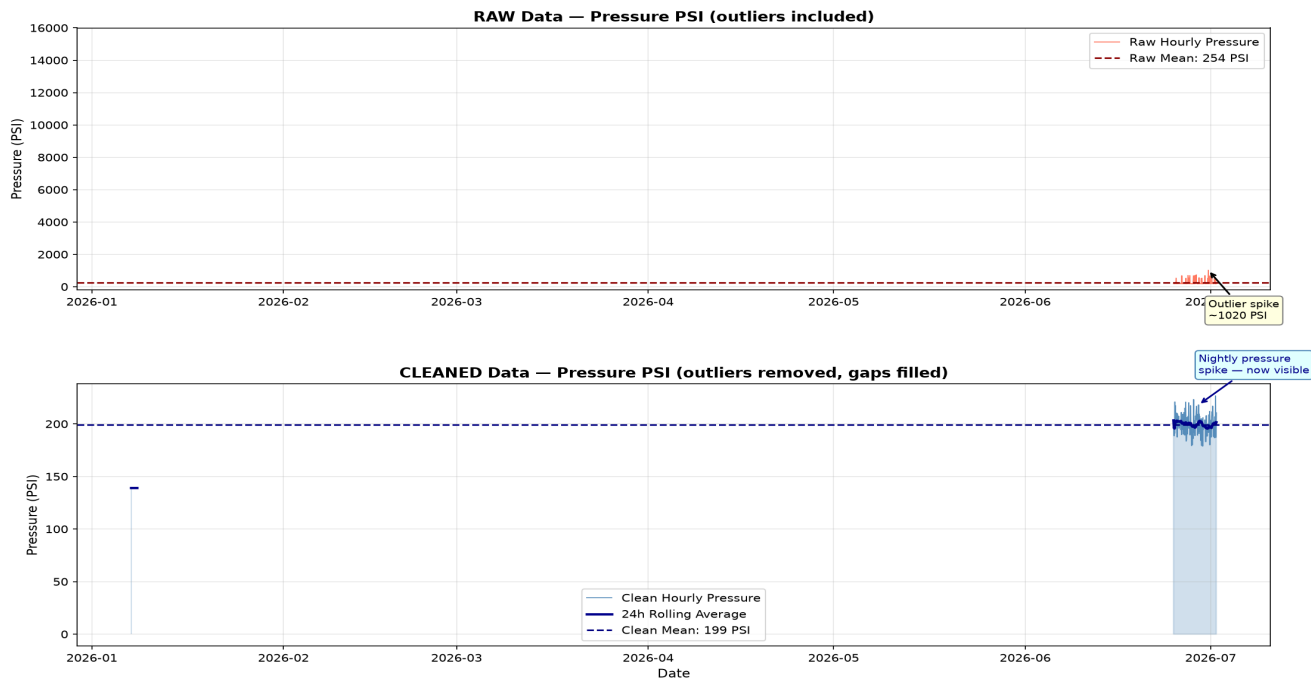


Week 2 Insight Report — Sensor Data Analysis

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Figure 1 — Raw vs Cleaned Pressure PSI



■ **TOP** — Raw data: pressure spikes reach 15,000 PSI due to faulty instruments. These swamp the real signal. The inflated mean (255 PSI) bears no relation to what the plant was actually doing.

✓ **BOTTOM** — Clean data: once the impossible readings are removed, a recurring overnight pressure rise (+20–30 PSI, midnight to 02:00) becomes clearly visible. This pattern was completely masked in the raw data.

Context — What was wrong with the data?

The sensor log arrived with five problems serious enough to make any analysis unreliable. Pressure readings ranged from **–50 PSI to 15,000 PSI** — values that are physically impossible for this plant and point straight to faulty instrumentation or data-entry errors. Temperature showed the same story, with entries at **–273 °C** (absolute zero) and **1,500 °C**. On top of that, the dataset contained **15 duplicate records**, almost certainly from double-logging at shift handovers. The Zone column was a mess — 15 different spellings for what should be 5 zones ('Zone_North', 'ZONE-NORTH', 'z_north' and so on), making any zone-level breakdown completely meaningless. Finally, around **1% of readings were missing entirely** across pressure, temperature, and flow rate. The combined effect was a dataset that looked noisy and unstable, with a mean pressure inflated to 255 PSI — 25% above the true operating level.

Insight — What did the clean data reveal?

After removing the impossible readings, dropping duplicates, standardising zone names, and filling gaps by interpolating between neighbouring readings, a clear operational pattern emerged that had been completely buried before: **pressure rises by 20–30 PSI every single night between midnight and 02:00**, consistently across the week. The raw data had suggested a stable, noisy system with no discernible trend. The clean data tells a different story — a systematic nightly cycle that repeats regardless of zone or day of the week. North Zone and West Zone show the sharpest elevation during this window. The 24-hour rolling average confirms this isn't random noise; it is a repeating pattern. The Night shift is operating at meaningfully higher pressures than Morning or Afternoon, and nothing in the current operating procedure accounts for it.

Action — What should we do about it?

Three things need to happen. **First, send maintenance into North Zone and West Zone this week during the midnight–02:00 window** to physically inspect what is causing the pressure rise. The most likely culprits are thermal pipe contraction as the plant cools overnight, valve control drift during low-demand hours, or reduced flow damping when production slows down. A walkthrough during the Night shift is the fastest way to confirm the cause. **Second, set an automated alert for any pressure reading above 230 PSI during Night shift** — the clean data shows the nightly peak regularly approaches this level, and an alert would flag any further escalation before it becomes a safety concern. **Third, the faulty sensors generating impossible readings must be repaired or replaced.** Until the instrumentation is reliable, the operations team is making decisions on corrupted data. Every night this goes unfixed is another night of flying blind during the highest-risk window in the operating cycle.